

A WORTH Subcase of Dalby's $RW(1, X) = 1$ Question

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Source and Question

This packet concerns Tim Dalby's paper *Properties of $R(X)$, $R(a, X)$ and $RW(a, X)$* , arXiv:2005.08492. In Section 5, page 13 of the arXiv PDF, Dalby asks:

“So it is an open problem: If either $R(1, X) = 1$ or $RW(1, X) = 1$ does X have $\text{Property}(M)$?”

PROPERTIES OF $R(X)$, $R(a, X)$ and $RW(a, X)$

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So it is an open problem: If either $R(1, X) = 1$ or $RW(1, X) = 1$ does X have $\text{Property}(M)$?

But in the mean time we can state the following propositions.

Proposition 5.3. *If X be a separable Banach space with $RW(1, X) = 1$ and sat-*

The result below gives a partial answer for the $RW(1, X) = 1$ half: the implication holds for WORTH spaces. It does not settle the unrestricted question and it does not address the weaker hypothesis $R(1, X) = 1$.

Definitions

Following Dalby, for $a > 0$,

$$RW(a, X) = \sup \left\{ \liminf_n \|x_n + x\| \wedge \liminf_n \|x_n - x\| : \|x\| \leq a, \|x_n\| \leq 1, x_n \rightharpoonup 0 \right\}.$$

A Banach space X has WORTH if for every weakly null sequence (x_n) and every $x \in X$,

$$\left| \|x_n + x\| - \|x_n - x\| \right| \longrightarrow 0.$$

$\text{Property}(m_\infty)$ is the identity

$$\limsup_n \|x_n + x\| = \limsup_n \|x_n\| \vee \|x\|$$

for every weakly null sequence (x_n) and every $x \in X$. It implies $\text{Property}(M)$.

Proof Intuition

The coefficient $RW(1, X)$ sees only the better of the two signs $x_n + x$ and $x_n - x$. This is exactly why the unrestricted question is hard: the coefficient may certify that one sign is small while the other sign carries the relevant obstruction. WORTH removes this ambiguity. Once the two signs have the same asymptotic size, the $RW(1, X) = 1$ estimate on a subsequence that realizes the limsup gives the missing upper bound, while the triangle inequality and weak lower semicontinuity give the matching lower bound.

Main Result

Theorem 1. *Let X be a Banach space with WORTH. If $RW(1, X) = 1$, then X has Property(m_∞). Consequently X has Property(M).*

Proof. Let (x_n) be weakly null and let $x \in X$. Put

$$r = \limsup_n \|x_n\|, \quad L = \limsup_n \|x_n + x\|, \quad a = r \vee \|x\|.$$

The case $a = 0$ is immediate, so assume $a > 0$.

First we prove the lower bound $a \leq L$. Since $x_n + x \rightharpoonup x$, weak lower semicontinuity gives $\|x\| \leq \liminf_n \|x_n + x\| \leq L$. Also

$$2\|x_n\| \leq \|x_n + x\| + \|x_n - x\|.$$

By WORTH, $\|x_n + x\| - \|x_n - x\| \rightarrow 0$, hence

$$r = \limsup_n \|x_n\| \leq L.$$

Thus $a \leq L$.

For the reverse inequality, fix $\varepsilon > 0$ and choose a subsequence (x_{n_k}) such that

$$\|x_{n_k} + x\| \rightarrow L \quad \text{and} \quad \|x_{n_k}\| \leq a + \varepsilon$$

for all k . Let $b = a + \varepsilon$ and set

$$y_k = \frac{x_{n_k}}{b}, \quad y = \frac{x}{b}.$$

Then (y_k) is weakly null, $\|y_k\| \leq 1$, and $\|y\| \leq 1$. WORTH applied to the subsequence gives

$$\|y_k + y\| \rightarrow \frac{L}{b} \quad \text{and} \quad \|y_k - y\| \rightarrow \frac{L}{b}.$$

Since $RW(1, X) = 1$,

$$\frac{L}{b} = \liminf_k \|y_k + y\| \wedge \liminf_k \|y_k - y\| \leq 1.$$

Therefore $L \leq a + \varepsilon$. Letting $\varepsilon \downarrow 0$ gives $L \leq a$.

We have proved

$$\limsup_n \|x_n + x\| = \limsup_n \|x_n\| \vee \|x\|$$

for every weakly null (x_n) and every $x \in X$, i.e. X has Property(m_∞).

Finally, if $\|u\| \leq \|v\|$, then for every weakly null (x_n) ,

$$\limsup_n \|x_n + u\| = r \vee \|u\| \leq r \vee \|v\| = \limsup_n \|x_n + v\|,$$

where $r = \limsup_n \|x_n\|$. This is Property(M). □

Corollary 1. *Any counterexample to the implication*

$$RW(1, X) = 1 \implies X \text{ has Property}(M)$$

must fail WORTH.

Verification and Novelty Notes

The proof uses only the definitions of $RW(1, X)$, WORTH, weak lower semicontinuity of the norm, and the triangle inequality. The normalization uses $a + \varepsilon$ rather than a so that the chosen subsequence satisfies the unit-ball requirement in the definition of $RW(1, X)$.

Bounded literature checks were performed for the phrases "RW(1, X) = 1" WORTH, "RW(1,X)=1" WORTH, "RW(a, X)" "Property(M)", and the source title. General web search and the local parsed arXiv corpus found the source paper and nearby Property(M)/WORTH discussions, but no later statement of this WORTH subcase.

Limitations

This is a partial result. It leaves open:

- whether $RW(1, X) = 1$ alone implies Property(M);
- whether $R(1, X) = 1$ implies Property(M);
- the corresponding dual variants discussed later in Dalby's paper.

References

- T. Dalby, *Properties of $R(X)$, $R(a, X)$ and $RW(a, X)$* , arXiv:2005.08492, 2020.
- B. Sims, *Orthogonality and fixed points of nonexpansive maps*, Proc. Centre Math. Anal. Austral. Nat. Univ. **20** (1988), 178–186.
- H. Rosenthal, *Some remarks concerning unconditional basic sequences*, Longhorn Notes, Texas Functional Analysis Seminar 1982–1983, University of Texas at Austin, 15–47.

Human Review Recommendation

Review as a likely valid partial answer. The key line to check is the subsequence normalization in the proof of the upper bound; after WORTH makes the two signs asymptotically equal, the $RW(1, X) = 1$ hypothesis applies directly.